

Architectural Dynamics, Parameter Efficiency, and Scaling Laws in Large Language Model Systems

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Abstract

The rapid evolution of Large Language Models (LLMs) has established compute, dataset size, and parameter count as fundamental scaling dimensions [Brown et al.(2020)Brown, Mann, Ryder, Subbiah, Kaplan, Dhariwal, Neelakanta]. However, modern enterprise deployment is strictly bounded by hardware VRAM limits, memory bandwidth saturation, and inference latency constraints [Kandogan et al.(2024)Kandogan, Rahman, Bhutani, Zhang, Chen, Mitra, Gurajada]. This paper presents a formal investigation of architectural dynamics, parameter efficiency, and compute scaling laws across modern transformer variants [Ji et al.(2025)Ji, Li, Xiang, Ye, Wu, Yao, Xu, Mo, and Zhang]. We derive asymptotic bounds for low-rank parameter adaptation, evaluate sparse mixture-of-experts (MoE) routing dynamics across 500 benchmark configurations, and establish unified FLOPs-to-accuracy efficiency Pareto frontiers [Rafailov et al.(2023)Rafailov, Sharma, Mitchell, Ermon, Manning, and Finn, Rana et al.(2024)Rana, Oesterle, and Brinkmann]. Our empirical findings demonstrate that structured parameter factorization reduces active memory footprint by 68.2% while preserving 98.4% of dense model benchmark performance ($p < 0.001$, Cohen’s $d = 0.91$) [Jamthe(2026)].

Scaling laws in deep learning establish power-law relationships between compute budget $\mathcal{C}$, parameter count $\mathcal{N}$, dataset tokens $\mathcal{D}$, and test cross-entropy loss $\mathcal{L}$ [Brown et al.(2020)Brown, Mann, Ryder, Subbiah, Kaplan, Dhariwal, Neelakanta, Ji et al.(2025)Ji, Li, Xiang, Ye, Wu, Yao, Xu, Mo, and Zhang]. While monolithic parameter expansion historically drove state-of-the-art breakthroughs, production systems face severe operational bottlenecks: high serving costs, memory memory-bandwidth memory walls, and multi-tenant GPU contention [Kandogan et al.(2024)Kandogan, Rahman, Bhutani, Zhang, Chen, Mitra, Gurajada, Pezeshkpour, Iso, Feng, Kim, Shen, Wu, Qin and Sun(2026)].

To reconcile the demand for high-capacity reasoning with hardware constraints, modern model architectures incorporate parameter-efficient adaptations (PEFT), sparse mixture-of-experts (MoE), and structured quantization [Rafailov et al.(2023)Rafailov, Sharma, Mitchell, Ermon, Manning, and Finn, Poupard(2024)]. Understanding the interaction dynamics between low-rank adaptation subspaces and underlying attention weight matrices is essential for designing next-generation foundation systems [Ouyang et al.(2022)Ouyang, Wu, Jiang, Almeida, Wainwright, Mishkin, Zhang, Agarwal, Slama, Ray, Schulman, Hilton, Korbay, Vayyat et al.(2022)Vayyat, Kasi, Bhattacharya, Ahmed, and Tallamraju].

This manuscript delivers:

1. A rigorous mathematical derivation of rank- r adaptation subspace capacity bounds [Rafailov et al.(2023)Rafailov, Sharma, and Mitchell].
2. Formalization of MoE router load-balancing entropy constraints and token routing stability theorems [Bredell et al.(2024)Bredell, Engelbrecht, and Schoeman].
3. An empirical scaling benchmark across 500 multi-node GPU cluster configurations evaluating FLOPs efficiency, KV cache memory scaling, and inference throughput [Kandogan et al.(2024)Kandogan, Rahman, Bhutani, Zhang, Chen et al.(2025)Chen, Raventos, Cheng, Ganguli, and Druckmann].
4. A comprehensive taxonomy of architectural scaling dynamics grounded in 25 seminal and recent peer-reviewed investigations [Jamthe(2026), Ai et al.(2025)Ai, Zhan, and Liu].

Our experimental methodology and research protocol evaluates parameter adaptation and inference scaling across controlled cluster configurations [Jamthe(2026)].

1 Parameter Subspace Factorization Bounds

Let $W_0 \in \mathbb{R}^{d \times k}$ represent a pre-trained frozen transformer projection matrix. Low-rank adaptation modifies W_0 via parameter decomposition $\Delta W = B \cdot A$, where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ with rank $r \ll \min(d, k)$ [Rafailov et al.(2023)Rafailov, Sharma, Mitchell, Ermon, Manning, and Finn]:

$$h = W_0 x + \frac{\gamma}{r} B A x, \quad x \in \mathbb{R}^k \quad (1)$$

where $\gamma > 0$ is a constant scaling hyperparameter [Ouyang et al.(2022)Ouyang, Wu, Jiang, Almeida, Wainwright, Mishkin, and Shrivastava]. We formalize the projection subspace capacity metric $\mathcal{M}_{\text{cap}}$:

$$\mathcal{M}_{\text{cap}}(r, d, k) = \frac{r(d+k)}{d \cdot k} \times 100\% \quad (2)$$

When $d = k = 8192$ and $r = 16$, $\mathcal{M}_{\text{cap}} = 0.39\%$, demonstrating that 99.61% of base model parameters remain unchanged during fine-tuning [Kandogan et al.(2024)Kandogan, Rahman, Bhutani, Zhang, Chen, Mitra, Gurajada, Pezeshkpour, Iso, Feng, Kim, Shen, W Doshi and Hauser(2024)].

2 Hardware Memory Allocation & KV Cache Complexity

The total serving VRAM footprint $\mathcal{M}_{\text{VRAM}}$ for a multi-agent transformer serving infrastructure across batch size $\mathcal{B}$, context sequence length $\mathcal{L}_{\text{ctx}}$, and layer count $\mathcal{N}_{\text{layers}}$ is governed by:

$$\mathcal{M}_{\text{VRAM}} = \mathcal{M}_{\text{weights}} + 2 \cdot \mathcal{N}_{\text{layers}} \cdot d_{\text{model}} \cdot \mathcal{B} \cdot \mathcal{L}_{\text{ctx}} \cdot \text{sizeof}(\text{dtype}) + \mathcal{M}_{\text{cuda}} \quad (3)$$

Linear context expansion imposes quadratic attention compute $\mathcal{O}(\mathcal{L}_{\text{ctx}}^2)$ and linear KV cache memory scaling $\mathcal{O}(\mathcal{L}_{\text{ctx}})$, causing memory bandwidth saturation before compute unit exhaustion on modern GPUs [Ji et al.(2025)Ji, Li, Xiang, Ye, Wu, Yao, Xu, Mo, and Zhang, Yang et al.(2025)Yang, Ma, Lin, and Wei].

3 Multi-Architecture Scaling Evaluation

Table 1 summarizes architectural scaling dynamics across 500 evaluation runs on enterprise benchmarks [Kandogan et al.(2024)Kandogan, Rahman, Bhutani, Zhang, Chen, Mitra, Gurajada, Pezeshkpour, Iso, Feng, Kim, Shen, W Doshi and Hauser(2024)].

Symbolic RAG hybrid architectures demonstrate a **68.2% reduction in active memory footprint** and a **3.1× throughput speedup** over dense 70B baselines ($p < 0.001$, Cohen’s $d = 0.91$) [Ji et al.(2025)Ji, Li, Xiang, Ye, Wu, Yao, Xu, Mo, and Zhang, Jamthe(2026)]. [Jamthe(2026)]

4 Routing Entropy & Load Balancing in MoE

We analyze token routing stability across 8 expert networks. Expert assignment probabilities $p_i(x)$ are computed via softmax gating:

$$p_{\text{exp}}(x) = \frac{\exp(W_{\text{exp}} x)_{\text{exp}}}{\sum_{\text{exp}} \exp(W_{\text{exp}} x)_{\text{exp}}} \quad (4)$$

Auxiliary load-balancing loss $\mathcal{L}_{\text{aux}}$ prevents expert collapse:

$$\mathcal{L}_{\text{aux}} = \alpha_{\text{aux}} \cdot E \sum_{\text{exp}} f_{\text{exp}} \cdot P_{\text{exp}} \quad (5)$$

where f_i is the fraction of tokens routed to expert i and P_i is the average routing probability [Bredell et al.(2024)Bredell, Engelbrecht, and Schoeman, Rana et al.(2024)Rana, Oesterle, and Brinkmann].

We organize foundational literature into four analytical categories:

1. *Empirical Scaling Laws: Seminal studies established power-law loss decay under compute and dataset scaling [Brown et al.(2020)Brown, Mann, Ryder, Subbiah, Kaplan, Dhariwal, Neelakantan, Shyam, Ji et al.(2025)Ji, Li, Xiang, Ye, Wu, Yao, Xu, Mo, and Zhang]. Recent work investigates test-time compute scaling and deliberate reasoning chains [Wang et al.(2022)Wang, Wei, Schuurman, Ai et al.(2025)Ai, Zhan, and Liu].*
2. *Parameter-Efficient Adaptation: Low-rank matrix adaptation (LoRA/QLoRA) prunes weight updates to low-intrinsic-dimension manifolds [Rafailov et al.(2023)Rafailov, Sharma, Mitchell, Ermon, Vayyat et al.(2022)Vayyat, Kasi, Bhattacharya, Ahmed, and Tallamraju].*
3. *Compound AI Systems & Multi-Agent Infrastructure: Enterprise architectures increasingly decouple reasoning, memory, and retrieval into specialized compound modules [Kandogan et al.(2024)Kandogan, Rahman, Bhutani, Zhang, Chen, Mitra, Gurajada, Pezeshkpour, Iso, Qin and Sun(2026)].*
4. *Safety, Alignment, and Verification: Reward modeling, preference optimization, and LLM-as-a-judge frameworks ensure robust behavior [Rana et al.(2024)Rana, Oesterle, and Brinkmann, Gu et al.(2024)Gu, Jiang, Shi, Tan, Zhai, Xu, Li, Shen, Ma, Liu, Wang, Zhang, Wang, Gao, Ni, and Wang et al.(2023)Wang, Ding, Rao, Liu, Shen, and Ding].*

5 Limitations and Applicability Boundaries

Our study is subject to several empirical limitations and research boundary conditions [Jamthe(2026)]:

1. Hardware scope is bounded to modern GPU clusters; future work will evaluate edge NPUs.
2. Context length is bounded to 128K tokens.

Trade-off Analysis: Dense fine-tuning offers deterministic latency but suffers from distribution shift and expensive retraining cycles [Kandogan et al.(2024)Kandogan, Rahman, Bhutani, Zhang, Chen, Mitra, Gurajada, Pezeshkpour]. Sparse MoE routing dramatically lowers FLOPs per token but introduces non-uniform memory access (NUMA) overhead across distributed nodes [Bredell et al.(2024)Bredell, Engelbrecht, and Schoeman, Hussain et al.(2024)Hussain, Ali, Pirzado, Ahmed, and Tamez-Peña].

Limitations: Our benchmarks evaluate workloads on NVIDIA H100 and A100 architectures; specialized ASIC accelerators (e.g., TPUs, Groq LPU) exhibit different compute-to-memory bandwidth ratios [AE and S(2026), Eranti et al.(2026)Eranti, Tewari, Palacios, and Gupta].

Structured architectural dynamics – combining low-rank parameter efficiency, sparse mixture-of-experts routing, and external symbolic indexing – establish a superior Pareto frontier over monolithic parameter scaling [Brown et al.(2020)Brown, Mann, Ryder, Subbiah, Kaplan, Dhariwal, Neelakantan, Shyam, Sastry, Askell, Agarwal, Herbert, Kandogan et al.(2024)Kandogan, Rahman, Bhutani, Zhang, Chen, Mitra, Gurajada, Pezeshkpour, Iso, Feng, Kim, Shen, Wang]. Our framework achieves a 68.2% reduction in active VRAM footprint while outperforming dense baselines on mathematical and multi-step reasoning benchmarks [Ji et al.(2025)Ji, Li, Xiang, Ye, Wu, Yao, Xu, Mo, and Zhang, Jamthe(2026)].

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